



TmelnikAPP

Future Development Roadmap & Technical Specifications



Current Project Status

85%

Core Features
Complete

7

Data Models

9

Services

8

Screens

3000+

Lines of Code

● 🔥 High Priority - Working Prototype

- ✓ Complete FeedbackScreen with categorized feedback collection **Planned**
- ✓ Implement NewsScreen with complete news system **Planned**
- ✓ Improve UI/UX with smooth animations and transitions **In Progress**
- ✓ Optimize performance and reduce loading times **Planned**
- ✓ Complete testing on real devices (iOS/Android) **Planned**

● ⚡ Medium Priority - Advanced Features

- ✓ Push notification system for project deadlines **Planned**
- ✓ Admin dashboard with statistics and metrics **Planned**
- ✓ Advanced search and filtering system for projects **Planned**
- ✓ Offline support with intelligent local caching **Planned**
- ✓ Automatic data backup system **Planned**

● ✨ Low Priority - Future Enhancements

- ✓ Internationalization (i18n) for multi-language support **Planned**
- ✓ Dark mode and theme customization **Planned**
- ✓ Advanced accessibility for users with disabilities **Planned**
- ✓ Integration with external systems (third-party APIs) **Planned**
- ✓ Advanced analytics and behavioral tracking **Planned**

● 🛠️ Technical Stack

Frontend

Flutter 3.13+
Dart 3.1+
Material Design 3

Backend

Firebase Auth
Cloud Firestore
Cloud Functions

Platforms

iOS • Android
Web • Desktop
Cross-platform

Tools

VS Code • Android Studio
Firebase CLI
Git



Development Timeline



Week 1 (Thursday)

Goal: Complete working prototype

- Complete FeedbackScreen
- Implement basic NewsScreen
- Testing on real devices
- Critical bug fixes



Week 2-3

Goal: Advanced features

- Push notification system
- Admin dashboard
- Search and filters
- Performance optimizations



Week 4+

Goal: Polish and improvements

- Dark mode
- Internationalization
- Accessibility
- Advanced analytics

● 🔒 Security & Performance

🛡️ Security

- Granular Firestore Security Rules
- Rate limiting for frequent operations
- Complete server-side validation
- Daily automatic backups

⚡ Performance

- Lazy loading for images
- Intelligent offline caching
- Firestore query optimization
- Asset compression

Success Metrics

<2s

Loading Time

99.9%

Uptime

<1%

Crash Rate

4.5+

User Rating

● Currently Implemented Features

Authentication System

- Google Sign-In integration
- Email/Password authentication
- Guest mode for browsing
- User role management (Admin/Normal)

Project Management

- Real-time project offers from Firestore
- Admin project creation/editing
- Instagram sharing functionality
- Project deadline tracking

Personal Diary

- Interactive calendar with TableCalendar
- Journal entries with mood tracking
- Automatic project event integration
- Firebase persistence

Architecture Overview

Models

ProjectOffer
UserRole
Feedback
News
JournalEntry

Services

UserRoleService
FirebaseFirestoreService
JournalService
ProjectService

Screens

AuthScreen
ProjectOffersScreen
DiaryCalendarScreen
AddProjectScreen

Utils

DebugLogger
RateLimiter
ProjectValidator
LoadingController

TmelnikAPP - Youth Exchange Management Platform

Developed with ❤️ using Flutter & Firebase

Repository: [GitHub](#)

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